

HISSAR, India

WMO: 421310

Lat: 29.1736N

Lon: 75.7393E

Elev: 700

StdP: 14.33

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 41.0	(c) 43.4	(d) 32.5	(e) 27.6	(f) 65.9	(g) 35.7	(h) 31.5	(i) 62.6	(j) 6.8	(k) 63.3	(l) 5.5	(m) 61.9	(n) 0.9	(o) 320	(p) 0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 20.7	(c) 112.0	(d) 74.7	(e) 109.5	(f) 74.9	(g) 106.8	(h) 75.1	(i) 84.3	(j) 95.7	(k) 83.5	(l) 94.7	(m) 82.8	(n) 93.6	(o) 3.1	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.8	169.9	89.2	81.0	165.5	88.6	80.4	162.1	88.1	49.4	96.4	48.5	94.7	47.7	93.8	95.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 5.5	(c) 5.2	(e) 38.2	(f) 116.0	(g) 3.6	(h) 1.6	(i) 35.6	(j) 117.2	(k) 33.5	(l) 118.1	(m) 31.5	(n) 119.1	(o) 28.9	(p) 120.3		
(a) 7.1	(b) 5.5	(c) 5.2	(e) 36.7	(f) 86.0	(g) 3.3	(h) 2.1	(i) 34.3	(j) 87.5	(k) 32.4	(l) 88.8	(m) 30.5	(n) 89.9	(o) 28.1	(p) 91.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.5	56.3	63.6	74.0	86.0	92.9	93.4	90.4	88.3	86.6	80.6	69.9	59.2
	DBStd	13.80	4.64	5.45	6.22	6.15	5.51	5.81	4.51	3.68	3.54	4.58	4.38	5.17
	HDD50	9	6	0	0	0	0	0	0	0	0	0	0	3
	HDD65	560	273	85	6	0	0	0	0	0	0	0	8	188
	CDD50	10412	201	380	743	1081	1330	1303	1253	1188	1098	950	597	288
	CDD65	5489	3	45	284	631	865	853	788	723	648	485	155	9
	CDH74	76390	16	364	2880	8886	13851	13661	11740	10124	8435	5139	1224	70
	CDH80	46931	0	72	1347	5700	9686	9466	7334	5740	4561	2672	351	2
(14) Wind	WSAvg	1.6	1.2	1.7	1.8	1.9	2.1	2.3	1.9	1.8	1.5	1.0	0.7	0.9
(15) Precipitation	PrecAvg	18.2	0.5	0.7	0.6	0.5	0.9	2.2	4.9	4.1	3.3	0.5	0.1	0.2
	PrecMax	30.4	2.0	3.5	4.0	3.9	3.3	6.9	18.7	10.2	9.1	5.6	0.5	1.1
	PrecMin	3.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0
	PrecStd	6.9	0.5	0.9	0.9	0.8	0.9	1.9	4.3	2.9	2.7	1.3	0.2	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.5	86.3	100.0	110.9	115.2	114.8	108.7	103.7	101.9	99.6	89.8	79.2
		MCWB	62.6	66.0	70.0	71.1	75.0	75.1	80.5	81.3	78.7	74.3	67.2	62.1
	2%	DB	71.9	81.7	95.0	107.8	112.6	111.9	105.4	100.9	99.7	97.0	86.8	76.0
		MCWB	58.6	64.0	69.3	70.3	73.9	76.2	80.8	81.3	78.3	74.2	66.4	60.7
	5%	DB	69.2	78.1	91.1	104.7	110.2	109.3	102.5	98.4	97.8	94.7	84.3	73.4
		MCWB	57.1	62.6	68.0	70.1	74.1	76.7	81.1	81.5	78.1	73.2	65.7	59.1
	10%	DB	66.5	74.9	87.6	101.4	107.4	106.3	99.7	96.2	95.8	92.3	81.4	70.6
		MCWB	55.5	61.2	67.0	69.4	73.7	77.5	81.5	81.7	78.3	72.2	64.8	57.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	68.9	74.4	77.2	82.3	84.3	85.3	85.1	83.7	80.4	70.4	65.1
		MCDB	72.9	80.8	90.2	100.8	104.5	100.5	95.7	95.0	94.0	93.2	84.0	74.6
	2%	WB	61.0	66.4	71.8	74.3	80.2	83.0	84.4	84.1	82.6	78.3	68.7	62.7
		MCDB	68.5	77.9	88.4	99.0	102.5	98.4	95.7	94.2	92.7	89.9	82.2	72.6
	5%	WB	59.1	64.6	70.1	72.5	78.5	82.0	83.6	83.4	81.6	76.5	67.3	60.8
		MCDB	66.0	74.9	86.2	96.9	100.3	96.8	94.8	93.1	91.3	87.9	80.5	70.4
	10%	WB	57.3	62.6	68.5	71.0	76.9	81.1	82.9	82.7	80.7	75.1	66.0	59.1
		MCDB	63.6	72.0	83.8	95.1	98.6	96.2	93.6	92.1	89.9	87.0	78.7	68.0
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	22.3	25.2	28.0	26.5	20.7	15.0	14.0	17.5	23.7	24.3	21.4
		MCDBR	23.7	25.4	29.1	30.4	28.9	25.0	18.6	17.3	20.5	25.2	26.4	25.2
		MCWBR	11.4	10.2	9.2	8.7	8.5	6.8	4.9	5.4	5.9	7.9	10.1	11.6
	5% WB	MCDBR	19.7	22.6	26.9	27.4	24.8	19.4	15.6	15.0	16.3	20.9	23.2	22.0
		MCWBR	11.0	10.4	10.3	9.8	9.2	6.6	5.3	5.5	6.0	8.1	9.8	10.9
(40) Clear-Sky Solar Irradiance	taub		0.485	0.463	0.492	0.568	0.668	0.748	0.766	0.655	0.572	0.644	0.606	0.533
	taud		1.852	1.968	1.924	1.778	1.619	1.521	1.522	1.733	1.867	1.614	1.643	1.756
	Ebn at Noon		227	251	253	239	216	198	194	215	229	197	189	204
	Edh at Noon		55	53	59	71	83	91	91	73	62	75	68	58
(44) All-Sky Solar Radiation	RadAvg		986	1374	1789	2083	2169	1929	1684	1689	1689	1489	1152	986
	RadStd		106	100	78	75	71	127	119	107	98	69	75	63

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page